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SOLVATED ELECTRON IN IRRADIATED MELTS OF ALKALINE HALIDES

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Abstract—This paper is a review of experimental results obtained by the authors while studying the formation and the properties of solvated electrons (e_s^-) in the melts of alkaline halides (NaCl, KCl, NaBr, KI, etc.) and their mixtures by pulse radiolysis method. The data on the optical absorption spectra of these species at different temperatures and their reactivity are considered. The conclusions on their nature are made. The review contains the data on molecular radical-ions X_2^- forming simultaneously with e_s^-. The possible mechanism of radiolysis of the systems under consideration is discussed.

INTRODUCTION

THE MELTS of alkaline halides are considered as perspective materials for nuclear reactor engineering.[1] Therefore the information on the mechanism of their radiolysis is of practical importance. Besides, the solvated electrons are formed during the action of ionizing radiation on the systems.[2,3] Obviously, the knowledge of the peculiarities of the behaviour of these species in ionic liquids at high temperatures (such systems are the melts of alkaline halides) is of great significance for physical chemistry of the electron in condensed phase. These circumstances stimulated us to carry out the systematic study of the properties of solvated electrons in irradiated melts of alkaline halides and their mixtures by pulse radiolysis method with optical registration. The results obtained during the study are summarized in the present paper. Some of these results were published in communications.[3–8]

EXPERIMENTAL

The linear electron accelerator U-12 (electron energy is 5 MeV, pulse duration is 2.3 μs, maximum dose per pulse is 12 krad) was used as a source of pulse-ionizing radiation. Fast spectrophotometric apparatus for the detection of transient species had the time resolution of 0.2 μs.[9] For the transformation of optical signal at wave-lengths less than 900 nm photomultiplier FEU-109 was applied; at longer wave-lengths (up to 1500–1600 nm) germanium photodiode was used. The spectral resolution was from 5 nm (in the region of 300–400 nm) to 15 nm (in the region of 800–1200 nm). The low limit of spectral measurements was 300 nm (for chloride melts), 350 nm (for bromide melts) and 400 nm (for iodide melts). It was due to the own absorption of the melts.

All salts used were "chemical pure" grade. They were purified in addition by recrystallization from bidistilled water. To remove the traces of oxygen and water the samples in quartz cells with plane-parallel walls were heated under vacuum at 300–400°C during 3 days. Then they were sealed and put into a cylindrical heater for melting and irradiation. The heater had windows for light and electron beams. The temperatures were measured by thermocouple and maintained within ±5°C. After melting the sample was irradiated by electron pulse.

RESULTS AND THEIR DISCUSSION

It was found that in all the melts studied the primary short-lived species of two types were formed as a result of the action of electron pulse.[3] The species of the first type absorb the light as a rule in red and IR parts of the spectrum. Their lifetime is about several microseconds. The species of the second type absorb the light in violet and UV regions of the spectrum. Their lifetime is considerably higher than the lifetime of the first type species. All the data obtained testify that the species of the first type are solvated electrons and the species of the second type are molecular radical-ions X_2^- (i.e. Cl_2^- in the melts of chlorides, Br_2^- in the melts of bromides and I_2^- in the melts of iodides). Let us begin the discussion from e_s^-.

Optical absorption spectra of e_s^-

The optical absorption spectra of e_s^- in the melts of NaCl, KCl and their mixtures at 800°C are shown as an example in Fig. 1.[5] The parameters of e_s^- spectra in all the melts studied are listed in Table 1. It is seen that optical absorption band is comparatively wide. Its half-width is equal to 0.9–1.7 eV (depending on the nature of the melt). The peak of the band is shifted to longer wave-lengths with the increase of the sizes of both cations and anions, i.e. at the transition from Li^+ to Cs^+ and from Cl^- to I^-.

In the case of LiBr melt the bands of e_s^- and Br_2^- were overlapped. The individual bands were

Table 1. Parameters of optical absorption spectra of e_s^- in melts of alkaline halides and their mixtures[a]

Melt [b]	Temperature, °C	λ_{max}, nm	$E_{\lambda max}$, eV	$W_{1/2}$, eV
LiBr	600	500	2.48	1.50
NaI	720	990	1.25	1.30
NaBr	800	945	1.31	1.49
60% NaBr + 40% NaCl	800	880	1.41	1.69
NaCl	800	730	1.69	1.67
80% NaCl + 20% KCl	800	790	1.57	1.61
60% NaCl + 40% KCl	800	820	1.51	1.50
50% NaCl + 50% KCl	800	870	1.43	1.45
40% NaCl + 60% KCl	800	900	1.38	1.32
20% NaCl + 80% KCl	800	975	1.27	1.20
KCl	800	1060	1.17	0.92
50% KCl + 50% KBr	800	1100	1.13	0.96
KBr	800	1110	1.12	1.00
KI	700	1130	1.10	1.60
RbCl	700	1120	1.11	1.10
66% CsCl + 34% NaCl	850	1000	1.24	1.00
66% CsCl + 34% NaCl	700	950	1.31	1.15
66% CsCl + 34% NaCl	550	900	1.38	1.15

a $E_{\lambda max}$ and $W_{1/2}$ are the energy corresponding to the maximum of the band and the half-width of the band, respectively.
b The composition of the mixtures is given in mole per cents.

separated on the basis of different lifetimes of the species. The absorption disappeared during 3 μs after the pulse was mainly due to e_s^- and the rest due to Br_2^-.

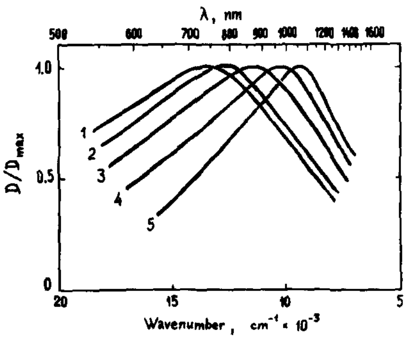

FIG. 1. Optical absorption spectra of e_s^- in melts of NaCl (1), KCl (5) and melts of their mixtures of composition (mole%): 80 + 20 (2), 50 + 50 (3) and 20 + 80 (4) at 800°C.

The following facts testify that the spectra under consideration belong to e_s^-. Firstly, the band intensity is considerably decreased (or it is not observed at all) when the melt contains the compounds which are electron scavengers. This is illustrated by Fig. 2, where the spectra in the melt of KBr (1) and in the melt of KBr containing 10^{-2} mole dm^{-3} Ca^{2+} (2) or 10^{-2} mole dm^{-3} Cd^{2+} (3) at 800°C immediately after the pulse are shown. As it is seen, the absorption in IR region in the presence of this amount of Cd^{2+} is practically not appearing. The same situation takes place in the cases when the melt contains the sufficient amounts of oxygen, ions Zn^{2+}, Tl^+, Ag^+ etc. On the basis of these results it is possible to conclude that the species under discussion is a reducing product of radiolysis.

Secondly, the experiments with the melts of the mixtures of alkaline halides show[5] that in these

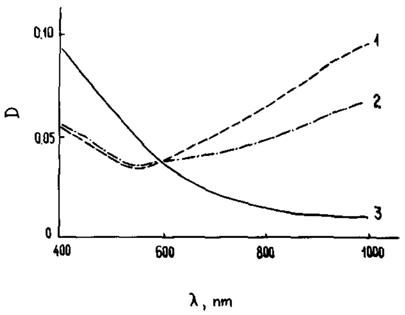

FIG. 2. Optical absorption spectra of melt of KBr (1) and melt of KBr, containing 10^{-2} mole dm^{-3} Ca^{2+} (2) or 10^{-2} mole dm^{-3} Cd^{2+} (3) at 800°C immediately after pulse.

systems the absorption band is uniform and it is monotonically changed during the variation of the melt composition. Besides, the half-width of the band is not more than the half-width of the band in any individual component of the mixture. It indicates that the band in the mixture is not the superposition of two unresolved bands. In this case if the absorbing species was atomic metal Me^0 the bands of each component of the mixture would be observed.

Thirdly, there is a peculiarity of optical absorption of e_s^- in the mixtures of the compounds which have similar nature (see, for example, Ref. 10), consisting of linear (or close to linear) dependence of the energy in the absorption peak on mole fraction of any component of the mixture. It was found[5] that it takes place also for the melts of the mixtures of alkaline halides. The linear dependence for the mixtures of NaCl and KCl at 800°C was obtained in Ref. 5. From the linear dependence it is possible to conclude that the composition of the nearest environment of the electron is proportional to the total composition of the melt.

Let us note that similar peculiarities of the e_s^- spectra in the melts of alkaline halides were also observed in the cases when this species is produced by dissolving the alkaline metal in the melt[11] or by electrochemical method.[12]

Nature of e_s^- in the melts.

Some conclusions on the nature of e_s^- can be made on the basis of comparative study of the radiolysis of alkaline halides in solid and liquid phases.[6] It is well-known that *F*-centers are formed under the action of ionizing radiation on the crystals. These centers are the electrons in anionic vacancies. Since the short range order in ionic liquid is remained it is possible to suppose e_s^- to be the species similar to *F*-centers. However, the transition of ionic crystals into melted state is accompanied[13] by the appearance of translation mobility of ions, the disappearance of the long range order and the sharp increase of specific volume of substance. Obviously, these factors should have an influence on the nature of electron centers in the melts. The consideration of these three factors shows that the most important factor respective to the electron centers during the transition from solid state to liquid one is the increase of specific volume. This process is connected particularly with the increase of the vacancies sizes.

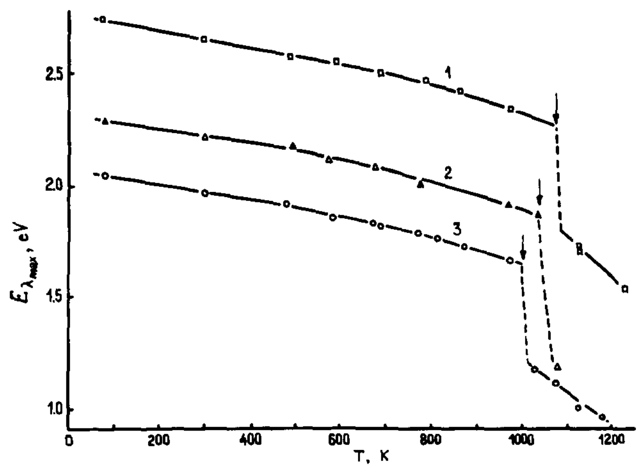

FIG. 3. Temperature dependence of $E\lambda_{max}$ for electron centers in monocrystals and melts of NaCl (1), KCl (2) and KBr (3). Arrows indicate melting points.

It should lead to the considerable shift of the peak of the optical absorption band of electron centers to longer wave-length in the melt as compared with the solid phase. Besides, because of the disordered melt structure the expanded vacancies should have nonuniform sizes. It should cause the considerable broadening of optical absorption band in the melt. The study[6] confirmed these conclusions.

The values of $E\lambda_{max}$ for electron centers in the monocrystals and the melts of NaCl (1), KCl (2) and KBr (3) at different temperatures are compared in Fig. 3. It is seen that the peak of optical absorption band of electron centers in the monocrystals is smoothly shifted to longer wave-length; however, near by melting point this shift is sharp.

The temperature dependence of $W_{1/2}$ for electron centers in NaCl (1), KBr (2) and KCl (3) is shown in Fig. 4. The dotted lines show the change of the half-width at the transiton to liquid state. As

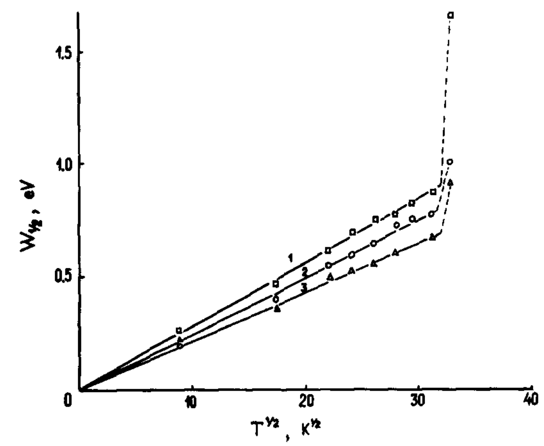

FIG. 4. Temperature dependence of $W_{1/2}$ for electron centers in monocrystals of NaCl (1), KBr (2) and KCl (3). Dotted lines indicate the variation of $W_{1/2}$ after transiton into liquid state.

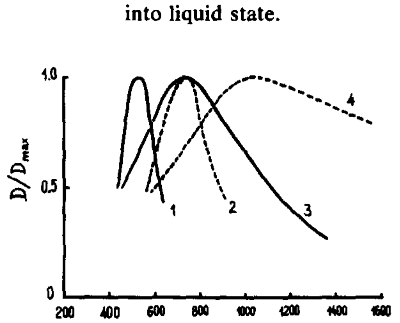

FIG. 5. Normalized optical absorption spectra of electron centers in NaCl (1, 3) and KBr (2, 4): 1, 2 are monocrystals at 700°C; 3, 4 are melts at 800°C.

it is seen, the half-width is considerably increased at this process.

Hence, on the basis of the study it is possible to conclude that solvated electrons in the melts are the electrons localised in expanded and nonuniform (in the size) anionic vacancies. From this it follows that the observed optical absorption spectrum of e_s^- may be considered as a sum of optical absorption bands of electron centers which are differed by the energy of the main optical transition. In this relation it may be proposed that the most high energetic absorption bands should belong to the electrons in vacancies corresponding to regular crystal lattice. Experimentally this point of view is confirmed by the absence of sharp change of short wavelength part of the spectra of electron centers at the transition from solid phase to liquid one. It is illustrated by Fig. 5, in which the normalised optical absorption spectra of electron centers in the monocrystals of NaCl (1) and KBr (2) at 700°C and their melts (3, 4) at 800°C are shown. As it is also seen from the figure, the sharp shift of the peak and the increase of the half-width at the transition through the melting point are mainly caused by the shift of the long wavelength boundary of the spectrum.

The microstructure of the electron trap in the melt seems to be differed from the structure in the crystal. It may be connected with the change of coordination numbers in the first (Me^+–X^-) and second (Me^+–Me^+, X^-–X^-) coordination spheres at the transition from solid to liquid phase. For example, coordination numbers in the first coordination sphere of the melts of NaCl and KCl are equal to 4.7 and 3.7, respectively, whereas in the crystals they are equal to 6.[13]

In addition, the conclusions made on the nature of e_s^- in the melts of alkaline halides are confirmed by the results of theoretical calculations performed in recent work.[14] In cited work the model in which electron localised in vacancy is considered to be surrounded by 4 cations interacting with anion-partners is used. In the model the anions are taken to describe the influence of the rest of the medium on energetics of the electron. Although the results of the calculation coincide with experiment uncompletely, they describe the general tendencies of the change of optical characteristics of the electrons at the transition from one melt to another.

Reactivity of e_s^- in melts

The decay of e_s^- even in the absence of the addition occurs very quickly (during 2–3 μs after the pulse). Because of it the quantitative measurements of e_s^- reactivity was carried out[4,7] from the dependence of the limit concentration of e_s^- during the pulse on the concentration of scavenger. The e_s^- concentration at the end of the pulse is close to the steady-state concentration even in the absence of scavenger. Hence

(1) $$d[e_s^-]/dt = G(I/100N) - V(e_s^-) - k_A[e_s^-][A] = 0,$$

where G is yield of e_s^-, I is dose rate, N is Avogadro's number, $V(e_s^-)$ is rate of reaction of e_s^- with matrix, A is scavenger and k_A is rate constant of reaction between e_s^- and A.

If we suppose that

(2) $$V(e_s^-) = k_I(e_s^-],$$

where k_I is rate constant of pseudomonomolecular decay of e_s^-, then

(3) $$[e_s^-]_s^0/[e_s^-]_s = 1 + k_A[A]/k_I,$$

where $[e_s^-]_s^0$ and $[e_s^-]_s$ are concentrations of e_s^- at the end of the pulse in the absence of scavenger and in its presence.

Using the optical densities D^0 (in the absence of A) and D (in the presence of A) we obtain

(4) $$D^0/D = 1 + k_A[A]/k_I.$$

The decay of e_s^- in pure matrix via pseudomonomolecular process can be explained by the occurrence of the reaction between e_s^- and cation of alkaline metal Me^+

(5) $$e_s^- + Me^+ \rightarrow Me$$

As it is seen from equation (4), the ratio D^0/D should be a linear function of scavenger concentration. It is observed in the experiment. Some dependencies of D^0/D on scavenger concentration is shown in Fig. 6. From the slopes of straight lines the ratios k_A/k_I are determined. Since k_I is obtained experimentally from the curve of e_s^- decay after the pulse in the absence of scavenger, then it is easy to find k_A from ratio k_A/K_I. The rate constants of reactions between e_s^- and different scavengers obtained by such a method are listed in Table 2.

The consideration of data given in Table 2 show that there are following peculiarities of e_s^- reactivity in the melts of alkaline halides. Firstly, the rate of e_s^- reactions depends on the nature of matrix. Among three investigated melts the highest reactivity of e_s^- is observed in KBr melt and the lowest one—in NaCl melt; KCl melt occupies the middle position. In this relation it is necessary to note that among investigated melts the most strong bond of the electron with medium is in the case of NaCl melt and the most weak one is in the case of

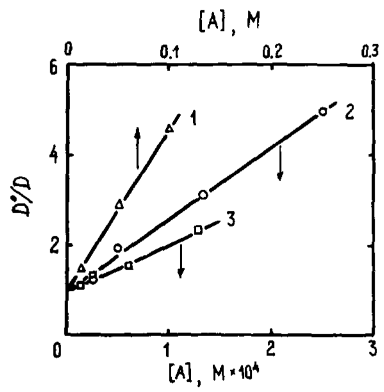

FIG. 6. Dependence of D^0/D on concentration of scavenger A in KBr melt at 800°C: 1, Ca^{2+}; 2, Cd^{2+}; 3, Zn^{2+}.

Table 2. Apparent rate constants of reactions of e_s^- in melts of alkaline halides[a]

Melt	k, $dm^3mole^{-1}s^{-1}$								
	Na^+	K^+	Ba^{2+}	Sr^{2+}	Ca^{2+}	Cd^{2+}	Zn^{2+}	Tl^+	Ag^+
NaCl (850°C)	$5x10^4$	-	-	-	$1x10^7$	$7x10^9$	$1.7x10^9$	$1.5x10^9$	$7x10^8$
KCl (800°C)	-	$7x10^4$	-	-	$2x10^7$	$1.3x10^{10}$	$2.8x10^9$	$2.3x10^9$	$1x10^9$
KBr (800°C)	-	$8x10^4$	$1x10^7$	$2.2x10^7$	$5.2x10^7$	$3x10^{10}$	$1.5x10^{10}$	$3.1x10^9$	$2x10^9$

[a] The error is ± 30% (such a value is due to k_I to be determined from very fast decay of e_s^- in pure melt)

KBr melt. It seems to be one of the main reasons of the dependence under discussion.

Secondly, e_s^- in the melts have comparatively high reactivity towards cations of alkali–earth elements. For example, the ratio of rate constants of reactions of e_s^- with ions Cd^{2+} and Ca^{2+} is equal to ~600 in KBr melt and ~700 in NaCl melt, whereas in water at room temperature these constants are differed by more than 5 orders of magnitude. It is very possible that higher reactivity of e_s^- towards cations of alkali–earth elements in the melts investigated is caused by insignificant effect of "solvation" of cations in the melts as compared with aqueous solutions.

Thirdly, although the melts are at very high temperatures, in all three melts the rate constant of e_s^- even with such an effective electron scavenger as ion Cd^{2+} is lower than the rate constant of this reaction in water at room temperatue. In 1975 the hypothesis that the effect under consideration is due to the reversibility of the reactions between e_s^- and scavengers in the melts at so high temperatures was spoken.[4] Recently this hypothesis was confirmed.[7]

The temperature dependence of the apparent rate constants of e_s^- reactions in the melts of the mixtures of NaCl (34 mole%) + CsCl (66 mole%) and LiBr (60 mole%) + KBr (40 mole%) was studied. These eutectic mixtures are melted at comparatively low temperatures (at 493 and 348°C). Because of it the experiments can be carried out within the sufficiently wide temperature interval. There were investigated the reactions of e_s^- with cations of matrix, ions Ca^{2+} and Cd^{2+} in NaCl–CsCl melt and with cations of matrix and ions Cd^{2+} in LiBr–KBr melt. The data obtained are shown in Figs. 7 and 8, respectively.

From the figures it is seen that the apparent rate constants k_{app} of e_s^- decay in pure matrices, i.e. the apparent rate constants of reactions between e_s^- and cations of alkaline metals, are increased with the increase of temperature up to 600–700°C. It is explained by normal activation nature of the process. However, at the further increase of temperature the decrease of k_{app} is observed. This decrease is more noticeable in the case of reactions with higher values of k_{app}. For example, in the case of Cd^{2+} ions the negative temperature coefficient of k_{app} takes place already at temperatures 400–500°C.

Such a dependence is explained by the occurrence of processes

$$e_s^- + Me^{n+} \rightarrow Me^{(n-1)+} \quad (6)$$

$$Me^{(n-1)+} \rightarrow Me^{n+} + e_s^- \quad (7)$$

and the total rate of e_s^- decay is determined by both reactions (6) and (7). In this case the apparent rate constant measured by usual way is expressed by equation

$$k_{app} = k_6 - k_7 \frac{[Me^{(n-1)+}]_s}{[Me^{n+}][e_s^-]_s}. \quad (8)$$

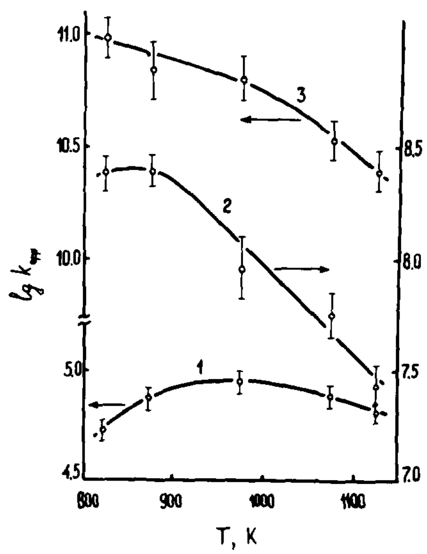

FIG. 7. Temperature dependence of apparent rate constants (k_{app}) for reactions of e_s^- with matrix (1), Ca^{2+} (2) and Cd^{2+} (3) in melt of mixture of composition 66 mole% CsCl + 34 mole% NaCl.

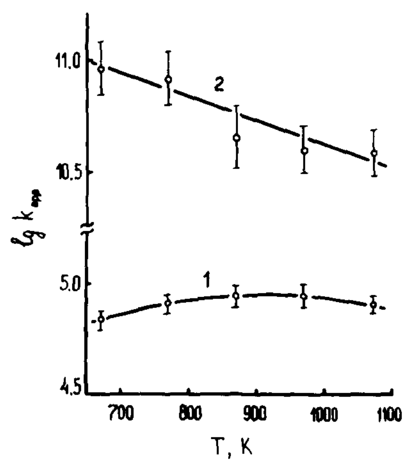

FIG. 8. Temperature dependence of apparent rate constants (k_{app}) for reactions of e_s^- with matrix (1) and Cd^{2+} (2) in melt of mixture of composition 60 mole% LiBr + 40 mole% KBr.

If the activation energy of reaction (7) is close to the energy of thermal dissociation of reduced metal ion $Me^{(n-1)+}$, i.e. it is considerably higher than the activation energy of reaction (6), then at comparatively low temperatures the second term in the right part of equation (8) is negligible and the temperature dependence of k_{app} is mainly determined by the temperature dependence of k_6. At sufficiently high temperatures the more fast temperature increase of k_7 in comparison with k_6 leads to the decrease of k_{app} with the increase of temperature; it is manifested in apparent negative activation energy of the process.

Hence, it is possible to speak on the high temperature limit of apparent rate constant at least for the reactions of reduction of metal ions. This maximum value is always lower than the diffusion limit of the actual rate constant of the respective reaction. It is caused by the reversibility of the reaction because of thermal dissociation of reduced metal ion.

The establishment of actual steady state via reactions (6) and (7) during the generation of e_s^- as a result of the action of ionizing radiation seems to be impossible because of the further reactions of $Me^{(n-1)+}$. For example, in the case of alkaline metal cations such reactions are the following (see also forthcoming discussions)

(9) $$Me^+ + Me \rightarrow Me_2^+,$$

(10) $$Me_2^+ + X_2^- \rightarrow 2Me^+ + 2X^-.$$

However, in the absence of oxidising species the equilibrium can be established as it takes place, for instance, in the solutions of alkaline metals in the melts of alkaline halides[1] where solvated electrons are comparatively stable.

Optical absorption spectra of X_2^-

The molecular radical-ions X_2^- are formed simultaneously with e_s^- under the radiolysis of the melts of alkaline halides.[3,8] The spectra of these species in the melts of KCl, KBr and KI are shown as an example in Fig. 9. The parameters of optical absorption spectra of X_2^- in the melts studied are presented in Table 3.[8] In the table the parameters of spectra of X_2^- in aqueous solutions at room temperature are also given for the comparison.[15,16] It is seen that Br_2^- and I_2^- in the melts, similar to aqueous solutions, have two bands: the main (at short wave-lengths) and the second (at long wave-lengths); Cl_2^- has only one band (at short wave-lengths). However, the ratio of molar extinction coefficients in maxima of the second and the main bands of Br_2^- and I_2^- in the melts is considerably higher than in aqueous solutions, i.e. the relative intensity of the second band in the melts is increased in the comparison with aqueous solutions. The same phenomenon was found earlier[17] for the spectrum of Br_2^- in the melt of tetrabutyl ammonium bromide where the ratio under consideration was equal to 0.2. Besides, the maxima of the bands in the melts of alkaline halides are shifted to longer wave-lengths in the comparison with aqueous solutions; the bands in the melts are more wide than in aqueous solutions. Obviously, the reason of these differences is that the melts are at high temperatures.

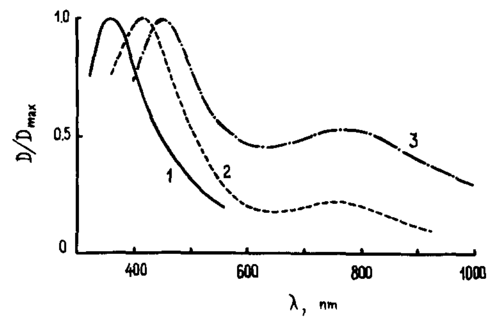

FIG 9. Optical absorption spectra of Cl_2^- (1), Br_2^- (2) and I_2^- (3) in melts of potassium halides: 1 is KCl (800°C); 2 is KBr (800°C) and 3 is KI (700°C).

Therefore, the following primary processes take place during the radiolysis of the melts of alkaline halides

(11) $$X^- \rightsquigarrow X + e^-,$$

(12) $$e^- \rightarrow e_s^-,$$

(13) $$X + X^- \rightarrow X_2^-.$$

Kinetics of decay of X_2^-[8]

The molecular radical-ions X_2^- decay independently of e_s^-, and the lifetimes of X_2^- are considerably longer than the lifetimes of e_s^-. The lifetimes of e_s^- in pure melts are equal to 2–3 μs, whereas the lifetime of Cl_2^- in the melts of chlorides under the same conditions is equal to ~10 μs, the lifetime of Br_2^- in the melts of bromides—to several tens of microseconds and the lifetime of I_2^- in the melts of iodides—to several hundreds of microseconds.

The decay of X_2^- is not accompanied by any change of its spectrum. It testifies to the absence of the superposition of the spectra of reaction products and gives a possibility to study the kinetics of X_2^- decay.

Table 3. Parameters of optical absorption spectra of X_2^- in melts of alkaline halides[a]

X_2^-	System [b]	Temperature, °C	λ_{MAX}, nm	$W_{1/2}$, eV	$\varepsilon_{II}/\varepsilon_I$ [c]
Cl_2^-	10^{-3} mole dm^{-3} Cl^-				
	in H_2O	25	340	0.93	-
	LiCl	700	360	1.1	-
	NaCl	850	365	1.2	-
	KCl	800	360	1.25	-
	RbCl	750	370	1.3	-
	CsCl	700	370	1.4	-
	66% CsCl + 34% NaCl	800	360	1.3	-
Br_2^-	10^{-3} mole dm^{-3} Br^-				
	in H_2O	25	360; 700	0.93 [d]	0.041
	LiBr	600	400; 725	1.15	0.2
	NaBr	760	420; 725	1.15	0.2
	KBr	800	420; 750	1.3	0.2
	40% KBr + 60% LiBr	700	400; 725	1.2	0.2
	40% KBr + 60% LiBr	850	420; 750	1.2	0.2
	CsBr	680	410; 750	1.2	0.2
I_2^-	10^{-3} mole dm^{-3} I^-				
	in H_2O	25	375; 725	0.8	0.27
	KI	700	450; 750	- [e]	0.5

[a] In the table the parameters of optical absorption spectra of X_2^- in aqueous solutions are included for the comparison[15, 16].

[b] Thecomposition of the mixtures is given in mole per cents.

[c] ε_I and ε_{II} are molar extinction coefficients in maxima of the main (at short wave-lengths) and the second (at long wave-lengths) bands, respectively.

[d] The values of $W_{1/2}$ for spectra of Br_2^- and I_2^- are given for the main bands.

[e] The calculation of $W_{1/2}$ in the case of I_2^- in the melts is difficult because of the noticeable superposition of two bands.

It was found that the decay of X_2^- in the dependence of the conditions occurs via three simplified mechanisms. The first mechanism takes place in the case of the melts not containing any additions. Taking into account the much longer duration of X_2^- decay in the comparison with the duration of e_s^- decay and the reversibility of the reaction between e_s^- and Me^+, it is reasonable to propose that X_2^- decays via reaction (10). Since the decays of X_2^- and e_s^- are separated in the time, it is possible to suggest that $k_9[Me^+] \gg k_{10}[X_2^-]$. From this it follows that initial concentrations of X_2^- and Me_2^+ should be equal. Hence, the decay of X_2^- should be expressed by equation

$$(14) \quad -d[X_2^-]/dt = -d[Me_2^+]/dt = k_{10}[Me_2^+][X_2^-] = k_{10}[X_2^-]^2.$$

On the basis of linear anamorphoses of X_2^- decay curves (as plots of $1/D$ vs t) the rate constants of reaction (10) shown in Table 4 were obtained. In the calculations of k_{10} the values of ϵ_{max} for X_2^- in aqueous solutions[15,16] were used; the corrections, taking into account the broadening of the bands in the melts and the redistribution of oscillator strength because of the increase of relative intensity of long wave-length band, were made.

From Table 4 it follows that the rate of reaction (10) is decreased during transition from Cl_2^- to Br_2^- and then to I_2^-, and the rate does not depend on temperature. The possible reason of the absence of temperature dependence is tunnel mechanism of electron transfer from Me_2^+ to X_2^-.

Let us emphasize that in the system under con-

Table 4. Rate constants of reaction (10) in melts of alkaline halides

X_2^-	Me	Melt [a]	Temperature, °C	k_{10}, $dm^3\ mole^{-1}\ s^{-1}$ [b]
Cl_2^-	Na	NaCl	850	2.0×10^{11}
	K	KCl	800	1.0×10^{11}
	Rb	RbCl	750	6.7×10^{10}
	Na(Cs)	34% NaCl + 66% CsCl	550	3.6×10^{11}
			600	4.0×10^{11}
			700	3.7×10^{11}
			750	4.0×10^{11}
			800	4.4×10^{11}
			850	4.8×10^{11}
Br_2^-	K	KBr	800	4.0×10^{10}
	Cs	CsBr	600	4.2×10^{10}
	Li(K)	60% LiBr + 40% KBr	400	2.4×10^{10}
			500	2.3×10^{10}
			600	2.9×10^{10}
			700	2.2×10^{10}
			800	2.2×10^{10}
			850	2.3×10^{10}
I_2^-	K	KI	700	3.8×10^{9}

[a] The composition of mixtures is given in mole per cents.

[b] The precision is ± 20%.

sideration the initial ions Me^+ and X^- are reformed via reactions (5), (9) and (10). In other words, the pure melts of alkaline halides because of this reformation should have the high radiation stability. In fact, the radiation chemical yield of Cl_2 at γ-radiolysis of the melts of chlorides is very low (<0.001 molec./100 eV).

The second mechanism is realized when the melt of alkaline halide contains cations of alkali-earth metals (for example, ions Ca^{2+}). In this case the curves of X_2^- decay consist of two parts: fast and slow decrease of the concentration. The analysis of these curves by means of analog computer shows that they are satisfactorily described by the reactions (below the example with Ca^{2+} is considered; Ca^+ in complex form is believed to react with X_2^-)

(15) $$e_s^- + Ca^{2+} \rightarrow Ca^+$$

(16) $$Ca^+ + X_2^- \rightarrow Ca^{2+} + 2X^-$$

(17) $$Ca^+ + Ca^+ \rightarrow Ca^0 + Ca^{2+}$$

(18) $$X_2^- + X_2^- \rightarrow X_3^- + X^-.$$

In reaction (15) the monovalent ions Ca^+ are formed; after transformation into complex form these ions interact with X_2^- (reaction (16)) and disproportionate (reaction (17)). Obviously, the role of reaction (17) is important when $k_{17} \gtrsim k_{16}$; in this case the slow reaction (18) can proceed to significant extent. Approximate values of rate constants of reactions (16)–(18) were determined for KBr melt containing Ca^{2+} (at 800°C). They are following: $k_{16} \approx k_{17} \approx 7 \times 10^{10}$ dm^3 mole^{-1} s^{-1} and $k_{18} < 10^9$ dm^3 mole^{-1} s^{-1}.

Hence, in the presence of ions of alkali-earth elements the increase of lifetime of X_2^- is observed. Let us note that, since ion X_3^- formed in reaction (18) then react with atomic metal appeared in reaction (17), the initial compound is also reformed in this system. However, because of the increase of lifetime of X_2^-, the further appearance of X_3^- and the partial evolution of halogen in gaseous phase the corrosion action of these systems towards the construction materials with which they can contact should be more effective.

The third mechanism is observed in the case when the melt contains the effective electron scavenger of type of Cd^{2+} or Zn^{2+}. In such systems X_2^- interacts with the product of the reaction between e_s^- and scavenger (for example, with ion Cd^+ if Cd^{2+} is added into the melt), i.e. it decays via reaction (as in the preceding cases reduced cation Cd^+ reacts with X_2^- being in complex form)

(19) $$Cd^+ + X_2^- \rightarrow Cd^{2+} + 2X^-,$$

Table 5. Rate constants of reactions between X_2^- and reduced cation of transiton metal in melts of alkaline halides[a]

X_2^-	Reduced cation	Melt	Temperature, °C	k, dm^3 mole^{-1} s^{-1}
Cl_2^-	Cd^+	NaCl	850	2.6×10^{11}
Br_2^-	Cd^+	KBr	800	7.0×10^{10}
Br_2^-	Zn^+	KBr	800	7.0×10^{10}
Br_2^-	Cd^+	60% LiBr + 40% KBr	400	3.6×10^{10}
			500	4.6×10^{10}
			600	6.4×10^{10}
			700	9.0×10^{10}
			800	1.0×10^{11}
			850	9.0×10^{10}

[a] The composition of mixture is given in mole per cents

[b] The precision is ±20%

the disproportionation of Cd^{+} being slow enough as compared with this reaction. The rate constants of some such reactions are listed in Table 5. It is seen that the rate of reaction (19) in the melt of mixture LiBr–KBr depends on temperature. From Arrhenius plot is was found that the activation energy of this reaction is about 25 kJ mole^{-1}. It is close to the activation energy of ions diffusion. Let us stress that in these melts, similar to other systems discussed, the initial compound is reformed.

CONCLUSION

The performed study of appearance and properties of short-lived e_s^- and X_2^- during irradiation of the melts of akaline halides by pulse radiolysis method gives possibility to explain the comparatively high radiation stability of these systems by the reformation of the initial compounds. At the same time this study shows that during radiolysis of the melts the various radiation chemical processes occur. It is especially characteristic for the melts containing the compounds which can be oxidized or reduced (for example, ions of transiton metals). This circumstance should be taken into consideration during the application of a melts of alkaline halides in all the fields of science and engineering in which ionizing radiation is used.

REFERENCES

1. V. L. Blinkin and V. M. Novikov, *Zhidkosolevye yadernye reaktory*, Atomizdat, Moscow, 1978.
2. S. J. Black and D. M. J. Compton, *J. Phys. Chem.* 1965, **69**, 4421.
3. I. E. Makarov, T. N. Zhukova and A. K. Pikaev, *Radiat. Eff.* 1974, **22**, 71.
4. I. E. Makarov, T. N. Zhukova and A. K. Pikaev, *Dokl. AN SSSR* 1975, **225**, 1103.
5. T. N. Zhukova, I. E. Makarov and A. K. Pikaev, *Izv. AN SSSR. Ser. khim.* 1977, 342.
6. I. E. Makarov, T. N. Zhukova and A. K. Pikaev, *Dokl. AN SSSR* 1978, **239**, 1144.
7. T. N. Zhukova, I. E. Makarov and A. K. Pikaev, *Dokl. AN SSSR* 1981, **261**, 409.
8. I. E. Makarov, T. N. Zhukova and A. K. Pikaev, *Izv. AN SSSR. Ser. khim.* 1982. To be published.
9. A. K. Pikaev, G. K. Sibirskaya, E. M. Shirshov, P. Ya Glazunov and V. I. Spitsyn, *Dok. AN SSSR* 1971, **200**, 383.
10. B. G. Ershov, I. E. Makarov and A. K. Pikaev, *Khim. vys. energy.* 1967, **1**, 404.
11. J. F. Rounsaville and J. J. Lagowski, *J. Phys. Chem.* 1968, **72**, 1111.
12. W. Schmitt and U. Schindewolf, *Ber. Bunsenges. Phys. Chem.* 1977, **81**, 584.
13. E. A. Ukshe, *Usp. khim.* 1965, **34**, 322.
14. C. W. Finley, N. R. Kestner and B. K. Rao, *Radiat. Phys. Chem.* 1980, **15**, 337.
15. L. I. Grossweiner and M. S. Matheson, *J. Phys. Chem.* 1957, **61**, 1089.
16. J. H. Baxendale, P. Sharpe and M. D. Ward, *Radiat. Phys. Chem.* 1975, **7**, 587.
17. I. E. Makarov, T. N. Zhukova and A. K. Pikaev, *Khim. vys. energy.* 1981, **15**, 402.